Cryptography and Network Security (BCS703) Question Bank

1. Explain the Symmetric Cipher Model with a neat diagram
2. Encrypt the message "HELLO" using Caesar Cipher with a shift of 3. Show the steps.
3. Explain the working principle of the Hill Cipher. How are matrix multiplication and key matrices used in encryption and decryption?
4. Describe the structure of the Data Encryption Standard (DES) and its major components.
5. Explain the principles and requirements of public key cryptosystems.
6. Using RSA, compute keys and encrypt the message M = 10 given *p = 13, q = 19, e = 5*.
7. Describe the Diffie–Hellman Key Exchange Algorithm with an example.
8. Write short notes on Elliptic Curve Cryptography (ECC) and its advantages over RSA.
9. Compare symmetric and asymmetric cryptography in terms of keys, speed, and applications.
10. Explain Pseudorandom Number Generators and describe any one (e.g., Linear Congruential Generator or Blum-Blum-Shub).
11. Define a cryptographic hash function. List its applications.
12. What are the limitations of these simple hash functions compared to modern cryptographic hash algorithms (like SHA-256)?
13. Differentiate between one-way hash functions and Message Authentication Codes (MAC).
14. List and briefly explain at least three applications of Hash Function.